

26-Mar-2024

Exercise 14.6

Triple Integral in cylindrical and spherical coordinates.

$$Q:1 \int_0^{2\pi} \int_0^1 \int_0^{\sqrt{1-r^2}} z r \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 r \left(\frac{z^2}{2} \right) \Big|_0^{\sqrt{1-r^2}} dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 \frac{1}{2} r (\sqrt{1-r^2})^2 dr \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^1 (r - r^3) dr \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left(\frac{r^2}{2} - \frac{r^4}{4} \right) \Big|_0^1 d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left(\frac{(1)^2}{2} - \frac{(1)^4}{4} \right) d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \frac{2-1}{4} d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \frac{1}{4} d\theta$$

$$= \frac{1}{8} \int_0^{2\pi} d\theta$$

$$= \frac{1}{8} (2\pi)$$

$$\Rightarrow \frac{1}{8} \left[\theta \Big|_0^{2\pi} \right]$$

$$\Rightarrow \frac{\pi}{4}$$

Converting Cartesian to cylindrical form.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$x^2 + y^2 = r^2$$

$$dV = dz dy dx$$

$$= r dz dr d\theta$$

Example 2:

Use cylindrical coordinates to evaluate the integral.

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx.$$

$$dz dy dx = r dz dr d\theta$$

$$r: 0 \rightarrow 3$$

$$\theta: 0 \rightarrow 2\pi$$

$$z: 0 \rightarrow 9 - x^2 - y^2$$

$$\because x^2 + y^2 = r^2$$

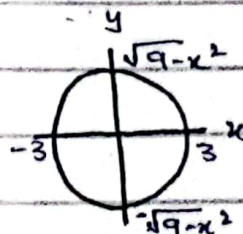
$$z: 0 \rightarrow 9 - r^2$$

$$\because x = r \cos \theta$$

$$y = \sqrt{9 - x^2}$$

$$x^2 + y^2 = 9$$

$$r = 3$$



$$= \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} r^2 \cos^2 \theta r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 r^3 \cos^2 \theta (z)_{0}^{9-r^2} dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 (9-r^2) r^3 \cos^2 \theta dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 (9r^3 - r^5) \cos^2 \theta dr d\theta$$

$$= \int_0^{2\pi} \left[\frac{9r^4}{4} - \frac{r^6}{6} \right]_0^3 \cos^2 \theta d\theta$$

$$= \int_0^{2\pi} \left[\frac{9(3)^4}{4} - \frac{(3)^6}{6} \right] \cos^2 \theta d\theta$$

$$= \frac{243}{4} \int_0^{2\pi} \cos^2 \theta d\theta$$

$$\because \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$= \frac{243}{4} \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{243}{8} \left(\theta + \frac{\sin 2\theta}{2} \right) \Big|_0^{2\pi}$$

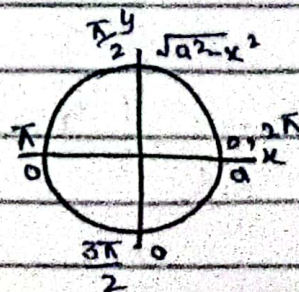
$$= \frac{243}{8} (2\pi + 0)$$

$$= \frac{243}{8} (2\pi)$$

$$= \frac{243 \pi}{4}$$

Q.17 $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2 dz dy dx$

$$dz dy dx = r dr d\theta$$



$$y = \sqrt{a^2 - x^2}$$

$$x^2 + y^2 = a^2$$

$$r^2 = a^2$$

$$r = a$$

$$z: 0 \rightarrow a^2 - x^2 - y^2$$

$$z: 0 \rightarrow a^2 - r^2$$

$$r: 0 \rightarrow a$$

$$\theta: 0 \rightarrow \pi/2$$

$$= \int_0^{\pi/2} \int_0^a \int_0^{a^2-r^2} r^2 \cos^2 \theta \cdot r \, dz \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^a r^3 \cos^2 \theta (z)_0^{a^2-r^2} \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^a r^3 \cos^2 \theta (a^2 - r^2) \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^a (a^2 r^3 - r^5) \cos^2 \theta \, dr \, d\theta$$

$$= \int_0^{\pi/2} \left(a^2 \frac{r^4}{4} - \frac{r^6}{6} \right)_0^a \cos^2 \theta \, d\theta$$

$$= \int_0^{\pi/2} \left(\frac{a^6}{4} - \frac{a^6}{6} \right) \cos^2 \theta \, d\theta$$

$$= \frac{a^6}{12} \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta$$

$$= \frac{a^6}{12} \times \frac{1}{2} \int_0^{\pi/2} 1 + \cos 2\theta \, d\theta$$

$$= \frac{a^6}{24} \left(\theta + \frac{\sin 2\theta}{2} \right)_0^{\pi/2}$$

$$= \frac{a^6}{24} \left(\frac{\pi}{2} \right)$$

$$= \frac{a^6 \pi}{48}$$

Example 1:

Use triple integration in cylindrical coordinates to find the volume of solid G that is bounded above by hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by xy -plane and by the cylinder $x^2 + y^2 = 9$.

$$V = \iiint_G dV$$

$$dV = r dz dr d\theta$$

$$z: 0 \rightarrow \sqrt{25 - x^2 - y^2}$$

$$z: 0 \rightarrow \sqrt{25 - r^2}$$

$$r = 3$$

$$r: 0 \rightarrow 3$$

$$\theta: 0 \rightarrow 2\pi$$

$$V = \int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 r (z) \Big|_0^{\sqrt{25-r^2}} dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 r (\sqrt{25-r^2}) dr d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \int_0^3 (-2r) \sqrt{25-r^2} \, dr \, d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \left(\frac{(25-r^2)^{3/2}}{3/2} \right)_0^3 d\theta$$

$$= -\frac{1}{3} \int_0^{2\pi} \left[(25-r^2)^{3/2} - (25)^{3/2} \right] d\theta$$

$$= -\frac{1}{3} \int_0^{2\pi} (64 - 125) \, d\theta$$

$$= \frac{61}{3} \int_0^{2\pi} d\theta$$

$$= \frac{61}{3} (\theta)_0^{2\pi} \Rightarrow \frac{61}{3} (2\pi)$$

$$= \frac{122}{3}$$

1-Apr-2024

Exercise 14.6

Spherical Coordinates

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$x^2 + y^2 + z^2 = \rho^2 : \text{eq of sphere}$$

ρ = radius of sphere

$$dz dy dx = \rho^2 \sin \phi d\rho d\phi d\theta$$

Use spherical Coordinates to evaluate the integral.

Example 4
$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2+y^2+z^2} dz dy dx$$

$$z = \sqrt{4-x^2-y^2}$$

$$z^2 = 4-x^2-y^2$$

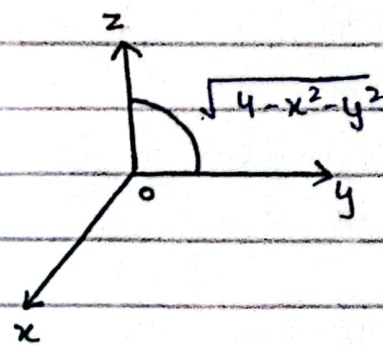
$$x^2 + y^2 + z^2 = 4$$

$$\rho^2 = 4$$

$$\rho = 2$$

$$\rho: 0 \rightarrow 2$$

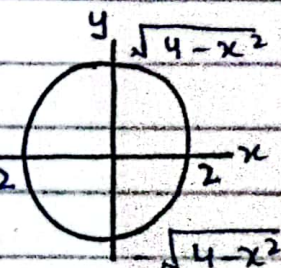
$$z: 0 \rightarrow \sqrt{4-x^2-y^2}$$



$$\phi = 0 \rightarrow \frac{\pi}{2}$$

$$\theta: 0 \rightarrow 2\pi$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^2 \cos^2 \phi \sqrt{\rho^2} \rho^2 \sin \phi d\rho d\phi d\theta$$



$$= \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^5 \cos^2 \phi \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \left(\frac{\rho^6}{6} \right)_0^2 \cos^2 \phi \sin \phi \, d\phi \, d\theta$$

$$= -\frac{2}{6} \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \phi (-\sin \phi) \, d\phi \, d\theta$$

$$= -\frac{64}{6} \int_0^{2\pi} \left(\frac{\cos^3 \phi}{3} \right)_0^{\pi/2} d\theta$$

$$= -\frac{64}{18} \int_0^{2\pi} \left(\cos^3 \left(\frac{\pi}{2} \right) - \cos^3 0 \right) d\theta$$

$$= -\frac{64}{18} (-1) \int_0^{2\pi} d\theta$$

$$= \frac{64}{18} (\theta)_0^{2\pi} \Rightarrow \frac{64}{18} \times 2\pi$$

$$= \frac{64}{9} \pi$$

Q: 20

$$\int_{-3}^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2} \, dz \, dx \, dy$$

$$z = \sqrt{9-x^2-y^2}$$

$$x^2+y^2+z^2 = 9$$

$$\rho^2 = 9$$

$$r = 3$$

$$r: 0 \rightarrow 3$$

$$z: -\sqrt{9-x^2-y^2} \rightarrow \sqrt{9-x^2-y^2}$$

$$\theta: 0 \rightarrow 2\pi$$

$$= \int_0^{2\pi} \int_0^{\pi} \int_0^3 r^2 \sin \phi \, dr \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \left(\frac{r^3}{3} \right)_0^3 \sin \phi \, d\phi \, d\theta$$

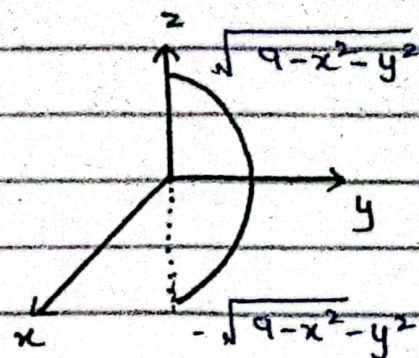
$$= \frac{3^3}{3} \int_0^{2\pi} (-\cos \phi)_0^{\pi} \, d\theta$$

$$= \frac{27}{3} \int_0^{2\pi} (-\cos \pi + \cos 0) \, d\theta$$

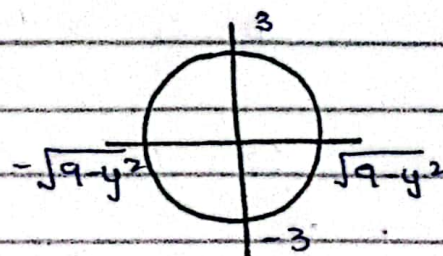
$$= \frac{27}{3} (1+1) \int_0^{2\pi} d\theta$$

$$= \frac{27}{3} (2) (\theta)_0^{2\pi}$$

$$= 81\pi$$



$$\phi: 0 \rightarrow \pi$$



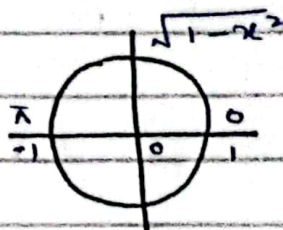
Q: 17 and 18 also included.

$$\text{Q: 18} \int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} e^{-(x^2+y^2+z^2)^{3/2}} dz dy dx$$

$$\rho: 0 \rightarrow 1$$

$$\phi: 0 \rightarrow \frac{\pi}{2}$$

$$\theta: 0 \rightarrow \pi$$



$$= \int_0^{\pi} \int_0^{\pi/2} \int_0^1 e^{-(\rho^2)^{3/2}} \rho^2 \sin \phi d\rho d\phi d\theta$$

$$= -\frac{1}{3} \int_0^{\pi} \int_0^{\pi/2} \int_0^1 e^{-\rho^3} (-3\rho^2) \sin \phi d\rho d\phi d\theta$$

$$= -\frac{1}{3} \int_0^{\pi} \int_0^{\pi/2} (e^{-\rho^3})_0^1 \sin \phi d\phi d\theta$$

$$= -\frac{1}{3} (e^{-1} - e^0) \int_0^{\pi} (-\cos \phi)_0^{\pi/2} d\theta$$

$$= -\frac{1}{3} (e^{-1} - 1) \int_0^{\pi} \left(-\cos \frac{\pi}{2} + \cos 0 \right) d\theta$$

$$= \frac{1 - e^{-1}}{3} (\theta)_0^{\pi} \Rightarrow \frac{(1 - e^{-1})}{3} (\pi)$$